A Small Sytematic Literary Review on agentic AI

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1 Details of the proposal

In this section, you will find key information about this project: all the specifications, the type of project carried out, and the range of points associated with it.

1.1 Full specification of the proposal

The main purpose of the proposal is to study what a Systematic Literary Review is and how the PRISMA methodology should be applied to develop a Systematic Literary Review.
It is recommended to review and understand an SLR in order to reproduce it. In particular, attention should be paid to the years of publication to be considered, the data sources, the queries used to search the resources, and the exclusion and inclusion criteria.
After fully understanding the theory and rules behind Systematic Literary Reviews, the project involves the development of a small Systematic Literary Review on agentic AI.

1.2 Proposal Type

The kind of proposal developed is "Standard."

1.3 The range of points of the proposal

The proposal has a medium level of difficulty, and the maximum number of points that can be earned with this project is 12.5.

Abstract

The rapid evolution of Large Language Models (LLM) has led to the emergence of Agentic AI, a term often misused in both academic and commercial contexts. This research presents a Systematic Literature Review (SLR) aimed at analyzing the nature of agency in LLM-based systems, with the aim of bridging the gap between classical agent theory and its modern implementations.

Our study followed the PRISMA guidelines and analyzed scientific articles published between 2024 and 2026 in the ACM Digital Library and IEEE Xplore databases. Starting from an initial set of 405 records, the screening process included 27 articles in the review after evaluating titles, abstracts, and inclusion and exclusion criteria. The quality assessment process selected the 10 articles with the highest scores to answer the main research questions. The results of the analysis show that Agentic AI is not a traditional autonomous agent, but rather a functional simulation. Agentic AI becomes proactive thanks to Agentic Workflows, which use multi-level planning, connect external tools, and include reflection loops. The study highlights a significant tradeoff: when moving from deterministic to probabilistic systems, Agentic AI gains flexibility, but challenges such as operational hallucinations, high latency, and the loss of mental state persistence arise.

The value of this work lies in providing a necessary taxonomy for emerging agentic paradigms, highlighting the structural shortcomings that impede true decision-making autonomy.

The research suggests that the best approach is to create hybrid architectures. Hybrid architectures would combine the adaptability of LLMs with the security and verification of formal systems, ensuring reliability in critical situations.

2 Introduction

The concept of "agent" in the field of Artificial Intelligence has very deep roots. Following the weak notion of Jennings and Wooldridge [4], an agent is a hardware or software-based computer system that is characterized by the following properties:

- Autonomy: it operates without direct human intervention and has control over its actions and internal states.
- Flexibility, which is distinguished by the following sub-properties:
  - Social ability: interacts with other agents or humans using an agent-communication language.
  - Reactivity: perceives the surrounding environment and responds to changes by taking actions to maximize success.
  - Pro-activity: not only responds to changes in the surrounding environment, but also exhibits goal-directed behavior and takes initiative on its own.
- Situatedness: it is equipped with sensors that perceive the environment.

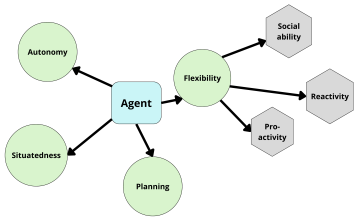

Figure 1: Weak notion of agent - Own elaboration

There is also the concept of "strong notion" which, in addition to assuming the same characteristics as "weak notion" (Figure 1), assumes that an agent is a system conceptualized or implemented using mentalistic and emotional concepts that are generally applied to humans.

The concept of "agentic AI" is relatively new. It should be noted that an agentic AI is not an agent, but attempts to simulate its life cycle by following the stages of Perceiving, Reasoning, Acting, and Learning. This simulation is increasingly achieved through an Agentic Workflow, in which the system's capabilities emerge not only from the model itself, but also from an iterative structure of planning, use of external tools and reflection. In this context, proactivity is not an intrinsic characteristic, but an emergent computational property derived from the context of the prompt. Unlike the strong notion, these systems do not possess persistent mental states, since their reasoning completely vanishes at the end of each session.
An agentic AI is monolithic and centralized; it relies on AI agents as fundamental components. These agents use Natural Language Processing (NLP) techniques, which belong to Large Language Models, to understand complex inputs, reason, and determine when to use external tools. Usually, an agentic AI uses an LLM that generates a pipeline and another LLM that executes it.

Given the increasingly widespread terminological confusion in recent years and the speed with which the term "Agentic AI" is spreading, a simple bibliographic search is no longer sufficient. This ambiguity has practical consequences: the lack of a clear taxonomy in these systems makes it difficult to assess their security, reliability, and robustness in critical situations. Therefore, a more specific and comprehensive research methodology is needed: We will therefore proceed with a systematic review of the literature (SLR). The ultimate goal of this review is to

summarize architectural trends and bridge the gap between classical agent theory and modern implementations, specifically addressing critical constraints such as latency and the risk of operational hallucinations [6].

3 Research Approach

In this section, we will explain the methodological approach used, which complies with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [11] guidelines.
There are literature reviews and general surveys focusing on the broad capabilities of Large Language Models and their various applications. However, this project involves a systematic study examining existing work and specific contributions provided in scientific articles published from 2024 to 2026 concerning the emerging paradigm of Agentic AI.
During the database search for this SLR study, most of the work on LLMs focuses on purely generative tasks or natural language processing. This project only considers those frameworks and specific contributions that address the emerging paradigm of Agentic AI.
When searching databases for this SLR study, most work on LLMs focuses on purely generative tasks or natural language processing.
This project considers only those frameworks and architectures that explicitly attempt to simulate the agentic lifecycle, that is perception, reasoning, and action, while recognizing that these systems are centralized simulations and not autonomous agents in the classical sense.
This article performs a small SLR based on the methodology inspired by PRISMA. The PRISMA methodology was chosen because it provides a standardized framework for creating and reporting systematic reviews and meta-analyses. The goal is to improve the transparency, replicability, and completeness of the entire analysis and review process.
The PRISMA methodology includes these 4 macro phases, also shown in (Figure 2):

1. Search criteria;
2. Inclusion/exclusion strategy;
3. Quality assessment;
4. Data collection and analysis.

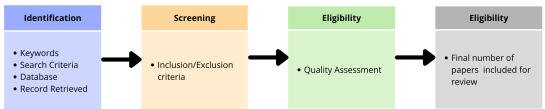

Figure 2: Methodology of the systematic review - Reworked from [5]

3.1 Research Question

In order to fully analyze the key concept of this research, A question has been defined that will guide the process and which we will try to answer:

> *"To what extent are current agentic AI systems based on LLMs capable of simulating the properties that characterize Agents as defined by Jennings and Wooldridge's weak notion?"*

Defining a main research question is essential in order to clearly distinguish the scientific concept of agent from modern commercial applications. As mentioned above, the main purpose of this SLR is to investigate the nature of agency in LLM-based systems, starting from the assumption that Agentic AI is not a classical agent, but a sophisticated functional simulation of one.

3.1.1 Relevance and Scomposition of the research

This question is crucially important because, unlike intelligent agents in traditional literature, which are often based on symbolic logic and Belief-Desire-Intention (BDI) architectures, Agentic AI stands out as a monolithic, centralized figure that simulates an agent's "Perception-Reasoning-Action" life cycle.
In order to better analyze how this simulation can be compared to agents, it was decided to divide the main research question into the following sub-questions:

- **RQ1**: How do agentic workflows simulate the goal-directed behavior typical of classical agents?
- **RQ2**: How is the perception of the environment and the dynamic response to changes managed through the integration of external tools? Does this simulation guarantee the same robustness as a traditional system?
- **RQ3**: What are the architectural shortcomings that prevent agentic AI from moving from simulated behavior to true autonomous decision making?

3.2 Searching Criteria

Based on the guideline of the PRISMA metodology, in order to select the scientific articles to best answer the sub-questions and consequently the research question, a specific search strategy was formulated [11]. The choice of keywords was dictated by the need to capture the emerging model of agentic AI, while simultaneously ensuring the right concentration on the reasoning capabilities that distinguish these systems from simple generative pipelines.
The search was carried out using Boolean operators (AND, OR) to create a structured query capable of including synonyms and related terminology.
The final search string used in the selected databases was:

("Agentic AI" OR "Agentic Workflow") AND ("Large Language Models" OR "LLM") AND "Reasoning"

In order to guarantee the correctness of the results, the search was carried out by applying the query to the following metadata fields:

- Title: to be sure that the main topic of the article was agent architectures.
- Abstract: to be able to check if the article dealt with simulation of the agent's life cycle (Perception, Reasoning, Action).
- Author keywords: to be able to identify that the studies were explicitly positioned in the field of LLM-based agents.

As specified in the research approach, a time filter was applied to limit the search to the years 2024 to 2026. This interval was chosen because the term "Agentic AI" has seen a surge in formal academic publications only in the last two years, making older literature less relevant for the analysis of current LLM-based simulations.

3.3 Data Sources

For this project, the databases used to search for articles are ACM Digital Library and IEEE Xplore.
Although there are numerous other sources available for automated research, we decided to use only these two portals for two main reasons: both, but more specifically the ACM Digital Library, are databases of scientific articles primarily focused on the collection and dissemination of academic knowledge in AI, machine learning, and related topics. These databases were initially selected to screen and filter articles that were not relevant to the primary research topic.

Database	URL
IEEE Xplore	`https://ieeexplore.ieee.org`
ACM Digital Library	`https://dl.acm.org/`

Table 1: Sources and Search Strings Used in This Study - Own elaboration

3.4 Inclusion and Exclusion Criteria

To ensure that the selected literature directly addressed the research question and maintained high academic standards, a set of formal inclusion and exclusion criteria was established. These criteria were applied during the screening and eligibility phases of the PRISMA process (Figure 4). The PRISMA Flow Diagram website was used to create and visualize the various phases of the PRISMA methodology [3]

3.4.1 Inclusion Criteria (IC)

An article was included in the final review if it met all of the following requirements:

- IC1 (Topic Relevance): The article focuses on "agentic AI" or "LLM-based agents" that explicitly implement a reasoning-action cycle.
- IC2 (Architectural Details): The study provides sufficient technical details regarding the system architecture
- IC3 (Simulation Agency): The work discusses how the system simulates at least one property of the weak notion (autonomy, reactivity, proactivity, or social ability).
- IC4 (Publication type): The article is a full research paper published in peer-reviewed journals or conference proceedings between 2024 and 2026.
- IC5 (Language and access): The article is written in English and is fully accessible through selected digital libraries.

3.4.2 Exclusion Criteria (EC)

Articles were excluded if they met any of the following conditions:

- EC1 (Focus exclusively on NLP): Articles focused exclusively on natural language processing (e.g., translation, summarization) without any component of "action" or interaction with the environment.
- EC2 (Classic agents): Studies on traditional symbolic agents (e.g., BDI, Jason, Prolog) that do not integrate modern large language models.
- EC3 (Non-academic material): Journal articles, posters, short workshop demos (less than 4 pages), or blog posts that do not have formal scientific validation.
- EC4 (Duplicates): Redundant records identified in both the ACM Digital Library and IEEE Xplore.
- EC5 (Open Access): Articles that are not open access.

An initial search returned 405 results (239 from ACM and 166 from IEEE); the IEEE search engine allows for search filters, while ACM does not.
By selecting the IEEE search filters, the results decreased from 166 to 18. After this initial screening, only results published from 2024 to 2026, which are entirely open access, remained. A further, non-automatic refinement based on abstract and title eliminated a further 11 results

from the selection.
Regarding the results obtained from ACM, relevant articles were manually excluded: of the 239 articles, 219 were discarded based on the type of result, and the accessibility of the document was already taken into account.
After this initial filtering phase, the 405 articles were reduced to 27 for evaluation in the next phase.

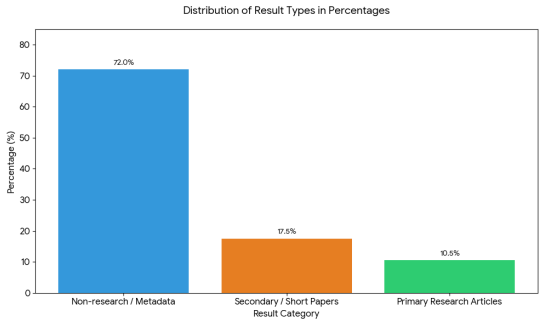

Figure 3: Distribution of Results Types - Own elaboration

The bar chart (Figure 3) shows the distribution of the types of results obtained. As can be seen, the majority of the results, equal to 72.0%, come from non-research sources or from metadata that are not relevant to the project's scope. This detail highlights the presence of noise within the context under study and, furthermore, justifies the need to apply a systematic and controlled process to filter the results obtained.

The second most significant portion, representing 17.5%, consists of articles that, while belonging to the correct type of searched results, do not necessarily offer new, original architectures and therefore do not represent the type of articles that can be included in the final selection.

The third and final type of results obtained, representing 10.5%, contains the academic articles that this project aimed to find; in fact, they are the academic research from which the papers to be used to complete this SLR will be extracted later.

3.5 Quality Assessment

Following the initial evaluation phase, which allowed for an initial screening of the papers found, we then proceeded to evaluate the quality.
This step is necessary to ensure that the selected papers have a satisfactory level of detail to answer the sub-questions and therefore the Research Question, thus avoiding work that is too superficial or solely based on hypotheses.

An evaluation system was used that assigns each paper a value based on four evaluation parameters. A rating system was used that assigns each article a value based on four evaluation parameters. The analysis was based on reading the abstract, discussion, and conclusions.
For each of these parameters a grade was assigned among the following values:

- 1 (Yes or Present)
- 0.5 (Partially Present)
- 0 (No or Absent)

The evaluation parameters considered are the following:

- QA1: Does the paper clearly explain how the system works internally?

- QA2: Does the study analyze or help us understand at least two of the following aspects: autonomy, reactivity, proactivity, or social skills?
- QA3: Does the article include real tests or benchmarks that demonstrate the effectiveness of the proposed simulation?
- QA4: Does the article discuss issues related to the technology?

Each paper can therefore obtain a maximum rating of four points. In order to select only the papers most consistent with the aim of this project, it was decided to consider a minimum threshold of 2.5 points; In fact, only and exclusively the documents that reached or exceeded this value were considered to be used for the purpose of answering the research questions.

The results of papers that have been validated using the quality assessment system are shown in the table 2 below.

Ref.	QA1 (Arch.)	QA2 (Weak N.)	QA3 (Valid.)	QA4 (Limit.)	Total
[7]	1.0	1.0	1.0	1.0	**4.0**
[1]	1.0	1.0	0.5	0.5	**3.0**
[16]	1.0	1.0	1.0	0.5	**3.5**
[10]	1.0	0.5	1.0	0.5	**3.0**
[15]	1.0	1.0	0.5	0.5	**3.0**
[12]	1.0	1.0	1.0	1.0	**4.0**
[14]	1.0	1.0	0.5	1.0	**3.5**
[13]	1.0	0.5	1.0	0.5	**3.0**
[9]	1.0	1.0	1.0	0.5	**3.5**
[17]	0.5	1.0	0.5	1.0	**3.0**

Table 2: Quality Assessment (QA) scores of the selected primary studies - Own elaboration

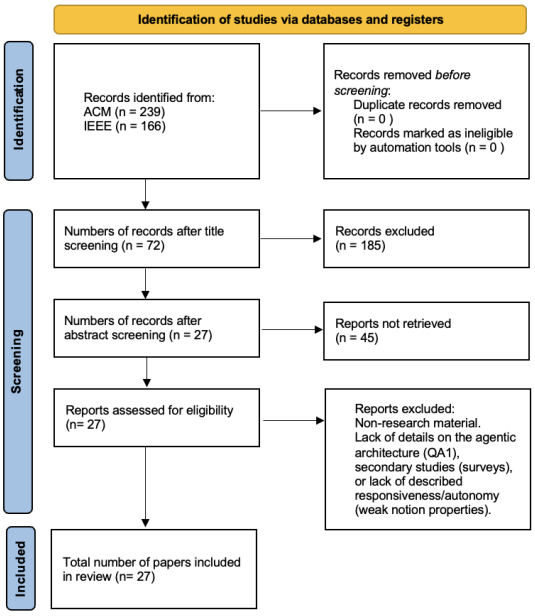

Figure 4: Flow chart. It shows the search and inclusion process - Own elaboration using the site: PRISMA flow diagram

3.6 Results

The use of the PRISMA protocol made it possible to carry out research to guarantee the completeness of the results obtained and made it possible to structure the investigation according to criteria of transparency and replicability. The screening process implemented also allowed us to filter out irrelevant or incomplete results, reducing the final dataset to 27 papers.

4 Results and Response to Research Questions

This section presents and discusses the results of the systematic literature review conducted according to the PRISMA protocol, described previously.
The primary objective of this chapter is to synthesize the collected evidence to provide comprehensive and scientifically grounded answers to the research questions formulated in the research plan.

The transition from raw data extracted from bibliographic databases to qualitative synthesis required a rigorous filtering and evaluation process to isolate the most significant contributions to the field of Agentic AI between 2024 and early 2026.
To accurately answer the research questions, the following analysis focuses on the 10 articles that received the highest scores in the quality assessment. These works were analyzed in their entirety to understand how architectures based on Large Language Models can simulate agentic behavior.

In the following subsections, the various research questions described above will be explored: first, will be analyzed the practical manifestation of Jennings and Wooldridge's "weak notion" properties (autonomy, reactivity, proactivity, and social skills) [4] within the frameworks implemented. Next, will be answered the second question by analyzing the recurring architectural patterns and functional modules (memory, planning, use of tools) that form the backbone of modern LLM-based agents. Finally, will be discussed the validation methodologies used by the authors and the challenges that remain in terms of security, latency, and reliability.

4.1 RQ1:How do agentic workflows simulate the goal-directed behavior typical of classical agents?

Proactivity represents the transition from a purely reactive system to one capable of goal-directed behavior. From the analysis of the ten selected studies, it emerges that this transition is not due to a will of the model, but rather to the implementation of agentic workflows (Figure 5 [8]) that orchestrate the LLM's reasoning.

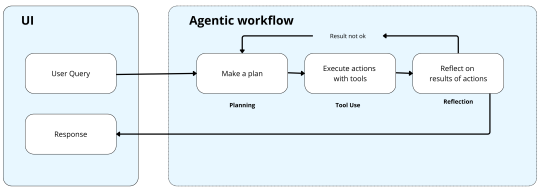

Figure 5: Agentic Workflow. (Reworked by Lollypop Design Studio [8])

4.1.1 Hierarchical planning and task breakdown

The first pillar of simulated proactivity is the ability to transform an abstract goal into a concrete action plan. In the article "XPF: An Agentic Workflow Framework for Business Process Automation" [13], the authors describe how an agent can receive a high-level goal and autonomously generate a sequence of complex activities. This ability to plan allows the agent

to constantly work towards the result, managing to imitate human initiative by planning the process through underlying steps without instructions for the intermediate user. Similarly, in "Agentic AI Systems: Architecture and Evaluation Using a Frictionless Parking Scenario"[3], agentic AI is defined as a computational entity oriented towards goals that eliminate the gap between reasoning and action with little or no human supervision. In his scenario, the system uses reasoning cycles to interpret a general intention (finding a parking space) in a series of autonomous maneuvers, ensuring that the goal is achieved despite environmental uncertainties.

4.1.2 Autonomous Initiative and Discovery

Proactivity also has the characteristic of being able to act without the need for specific requests. An excellent example of this can be found in "AutoMind: Automated Insight Discovery via Multi-Agent Navigation" [15], where agents proactively explore large data sets to discover meaningful patterns. Rather than responding to isolated queries, the agent decides to take the initiative and explore data paths based on an internal logic of information usefulness. This behavior can also be seen in "LLM-based agentic workflow on verbal and non-verbal audiovisual perceptions and actions for proactive situated human-robot interactions" [14]. This other system also doesn't simply react to a user's voice command, but rather proactively decides when to take the initiative and initiate the interaction based on how the environment is perceived, demonstrating situated proactivity.

4.1.3 Team Orchestration and Coordination

Proactivity can also manifest itself as the ability to anticipate project needs and manage resources. In "Orchestrating Human-AI Teams: The Manager Agent as aUnifying Research Challenge"[9], the introduction of a Manager Agent shows a system that proactively monitors the progress of a joint activity and assigns subtasks to team members, whether human or digital. This is not a passive response, but a supervisory activity aimed at ensuring the efficient achievement of the goal. Furthermore, the framework presented in "DAWN: Designing Distributed Agents in a Worldwide Network" [1] allows agents to proactively search for partners or services within a network to solve subproblems, demonstrating an initiative that extends beyond the boundaries of a single local system.

4.1.4 Persistence and error recovery

Classic proactivity requires persistence. In "RALLY: Role-Adaptive LLM-Driven Yoked Navigation for Agentic UAV Swarms"[16], the agentic workflow allows a swarm of drones to proactively adapt their roles in the event of a member error or changes in the environment. Instead of stopping and waiting for a new command, the system reprograms itself to ensure mission success. This persistent pursuit of a goal is what is defined in "INSYTE: A Classification Framework for Traditional to Agentic AI Systems" [12] as a high level of agency: the transition from a system that follows fixed rules to one that dynamically manages its own workflow to fulfill an intention. However, as noted in "Towards Safe Agentic AI Performance Engineering"[17], this proactive behavior must be contained within safety limits to prevent autonomous actions from causing system instability.
Agentic workflows simulate proactivity by shifting the LLM from a reactive question and answer engine to a planning and execution engine.
Using hierarchical planning, autonomous exploration, and adaptive coordination, these systems maintain a goal-oriented trajectory that closely aligns with the classical definition of a proactive agent.

4.2 RQ2: How is the perception of the environment and the dynamic response to changes managed through the integration of external tools?

Responsiveness, i.e. the ability to perceive and respond to environmental changes, is simulated in agentic systems by connecting LLM reasoning with external data sources and execution modules. Unlike traditional systems where perception is hard-coded, agentic workflows use tools to ground model logic in real or digital contexts.

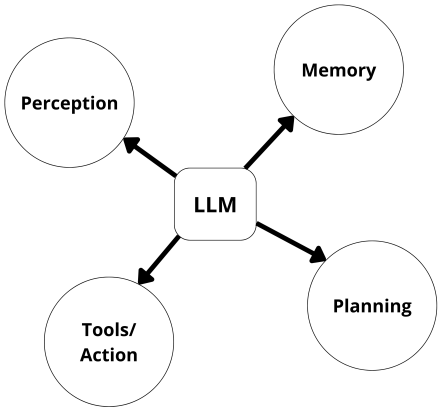

Figure 6: LLM-as-a-Brain Architecture - Own elaboration

4.2.1 Multimodal perception and Situatedness

In systems that interact with the physical world, perception is handled by specialized wrappers that translate raw sensor data into a format that can be processed by the LLM. Sucal et al. [14] demonstrate, in "LLM-based agentic workflow on verbal and non-verbal audiovisual perceptions and actions for proactive situated human-robot interactions", how an agent perceives the environment through audiovisual inputs. By integrating tools that process verbal and non-verbal signals, the workflow allows the robot to dynamically adapt its response based on the user's presence and tone. Similarly, Wang et al. [16] in "Rally: Role-adaptive LLM-driven yoked navigation for agentic UAV swarms" describe how drones perceive their relative position and environmental obstacles. The LLM acts as a high-level controller (Figure 6) that receives processed sensor data as context and uses navigation tools to adjust the swarm's trajectory in real time.

4.2.2 Dynamic response via knowledge integration

Perception is not limited to physical sensors; it also includes the ability to interpret complex information environments.
The authors of "Agentic reasoning for social event extrapolation: Integrating knowledge graphs and language models" [10] show how an agent perceives social trends by querying Knowledge Graphs. Knowledge Graphs serve as an external tool that provides a structured environmental context, allowing the agent to generate a dynamic response to social changes that a static LLM could not detect.
In the business environment, on the other hand, "Xpf: Agentic AI system for business workflow automation" [13] is described as managing environmental changes, such as data updates or execution errors, using tools that monitor the status of the workflow. If a tool returns an error, the workflow perceives this change and replans the next steps.

4.3 Does this simulation guarantee the same robustness as a traditional system?

The transition from deterministic reactive systems to probabilistic agentic systems introduces a trade-off between flexibility and reliability.
The analyzed papers suggest that, although Agentic AI excels in managing unexpected scenarios, it does not yet guarantee the same level of robustness as traditional hard-coded systems.

Characteristic	Traditional Systems	Agentic AI Systems
Basic Logic	Rigid, based on "If-Then" rules.	Flexible, based on goals (*Goal-oriented*).
Predictability	**High**: The output is constant and deterministic.	**Variable**: Stochastic and probabilistic nature.
Flexibility	**Low**: Requires manual reprogramming.	**High**: Dynamically adapts to changes [12].
Error Handling	Handled via predefined exceptions.	Handled via self-reflection loops.
Main Risk	Software bugs or unhandled inputs.	**Hallucinations** or illogical actions [8].
Robustness	Structural and mathematically verifiable.	Functional; requires monitoring [13].
Latency	**Minimal**: Instantaneous execution.	**High**: Due to LLM reasoning [10].

Table 3: Comparison between traditional deterministic systems and LLM-based agentic systems - Own elaboration

4.3.1 Non-deterministic risks and lack of reliability

Traditional reactive systems follow an "if-then" logic that ensures predictable results. Agentic AI, however, relies on outputs from LLMs which can prove inconsistent. In "Insyte: A classification framework for traditional to agentic AI systems" [12], the authors explicitly discuss this challenge, noting that as agency increases, so does the difficulty of verifying system behavior. Simulating reactivity via LLM can lead to "hallucinations in action," where the agent correctly perceives a change, but chooses an illogical tool or sequence of actions. Khamis [7], in his evaluation of the "frictionless parking scenario", also points out that, although agentic systems are more adaptable to complex urban contexts than static algorithms, they require much more rigorous testing to meet the safety standards of traditional automotive software.

4.3.2 Latency and performance instability

A major obstacle to robustness is the latency inherent to LLM reasoning. As noted by Sucal et al. [14], the time required for the LLM to process multimodal perceptions can compromise the fluidity of real-time interaction, a problem that traditional low-latency reactive systems do not present. Furthermore, in "Towards Safe Agentic AI Performance Engineering", edited by Williams and the other authors [17], it is highlighted that agent systems can exhibit unpredictable spikes in resource consumption. According to them, because these systems are not inherently robust, they must be "contained" using low-level system tools to monitor and limit autonomous actions in real time. This suggests that the current state of Agentic AI requires a hybrid approach. Rugged traditional monitors need to be used to supervise the agentic "brain". This is flexible but unstable.

4.3.3 Role adaptation and fault tolerance

Despite these weaknesses, agentic systems offer a new type of functional robustness through adaptation. As shown in “Rally: Role-adaptive llm-driven yoked navigation for agentic uav swarms” [16], if a component fails, the agentic workflow can proactively reassign roles within a swarm, which is a level of resilience that a traditional system with fixed roles would not have. However, as highlighted in “Automind: Automated insight discovery via multi-agent navigation” [15], this robustness strongly depends on the quality of multi-agent coordination; if the communication protocol or LLM reasoning fails, the entire discovery process may stall.

4.4 RQ3: What are the architectural shortcomings that prevent agentic AI from moving from simulated behavior to true autonomous decision making?

Despite the effectiveness of agent-based workflows, the literature highlights that current autonomy is an advanced form of context-driven simulation rather than intrinsic decision-making will. Analysis of the primary studies led to the identification of the following key gaps.

4.4.1 Absence of explicit mental states (BDI Model)

One of the main limitations lies in the difference between the architecture of LLMs (Figure 8 and classic models of agency, such as Belief-Desire-Intention (Figure 7). According to Porter and other authors [12], current systems do not have mental states that are persistent and independent of the immediate context, as described in ”Insyte: A classification framework for traditional to agentic ai systems” . While a classical agent has stable beliefs and desires, the LLM-based agent constructs its intention probabilistically, based on the current prompt. Therefore, an architecture that separates knowledge of the world (Beliefs) from long-term objectives (Desires) is missing, making the agent’s decision dependent on the quality of the formulation of the prompt rather than on a constant internal logic.

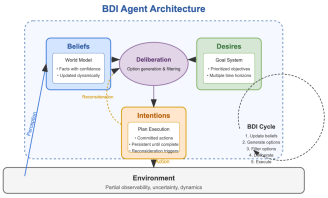

Figure 7: BDI model - Adapted from: The Belief-Desire-Intention Model [2]

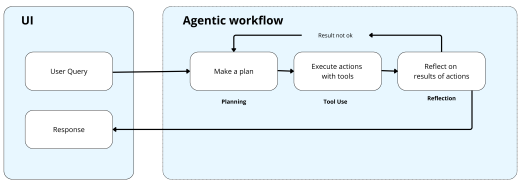

Figure 8: Agentic Workflow - Own elaboration

4.4.2 Presence of an intrinsic dependence on external stimuli

The simulated agency depends entirely on user input to activate the reasoning loop. When describing the “Xpf: Agentic AI system for business workflow automation” system, Rao et al. [13] clarify that the workflow starts only after a “high-level objective” is injected from outside. In the absence of this stimulus, the agent has no intrinsic motivation to act. This gap highlights that the agent does not possess self-triggering autonomy: it is a very high-level reactive system that breaks down the objectives of others, but is not capable of generating them autonomously, a fundamental requirement for true decision-making autonomy.

4.4.3 The gap between retrieving information and understanding the world

While agents use external tools to fill their knowledge gaps, this process remains a form of "consultation" and not "learning." In "Agentic reasoning for social event extrapolation: Integrating knowledge graphs and language models" [10], the authors show how the integration of Knowledge Graphs (KGs) is necessary to provide the agent with a structure of the world that the LLM does not intrinsically possess. However, the agent does not understand the graph, but uses it as an external database. This architectural gap indicates that agency is limited by situated knowledge that vanishes once the session ends, preventing the development of an experience that can autonomously guide future decisions.

4.4.4 Stochastic Nature vs. Decision Determinism

A critical obstacle to true autonomy is the nondeterministic nature of LLMs. In their paper "Towards Safe Agentic AI Performance Engineering," Williams and colleagues [17] emphasize that we cannot yet "trust" the agent's decisions, as the decision-making process is a probabilistic black box. In a real autonomous system, the decision must be verifiable and consistent. In contrast, Agentic AI may make different decisions under the same environmental conditions due to model variance. According to the authors, this gap requires a hybrid architecture in which an external control system must constantly monitor the agent and confirm that the current autonomy is granted and monitored, not intrinsic.

4.4.5 Limitations in complex multi-agent coordination

Finally, true autonomy would require complex social understanding. Masters and colleagues [9] in "Orchestrating Human-AI Teams: The Manager Agent as a Unifying Research Challenge" point out that, although an agent can act as a manager, its coordination capacity is limited by the lack of a theory of mind regarding other team members. The agent simulates coordination based on management language models, but lacks an understanding of deep human dynamics, which prevents it from making autonomous decisions that take into account the needs and emotional capacities of its collaborators.

5 Discussion and Findings

This section summarizes the findings from the analysis of the ten selected studies to provide a general overview of the capabilities and critical issues of Agentic AI. The goal, therefore, is to go beyond the simple description of data to interpret the ongoing shift in approach: the transition from deterministic software systems to probabilistic systems that, while not actual agents possessing a will of their own, simulate autonomous behavior through complex architectures.

5.1 The trade-off between flexibility and robustness

The comparative analysis clearly highlights that adopting agentic AI currently represents a significant compromise. Traditional systems operate according to a deterministic "If-Then" logic. This approach guarantees absolute predictability and significant structural robustness, but it results in an almost total loss of flexibility in the event of unforeseen situations. On the contrary, systems based on Agentic AI invert this paradigm: they offer a high capacity for dynamic adaptation to environmental changes, but introduce a stochastic and probabilistic nature which makes the output variable and not always predictable. The main risk therefore shifts from classic software bugs to so-called "hallucinations", i.e. the system can undertake illogical sequences despite starting from correct data.

5.2 Agency Simulation: From the BDI Model to Workflows

A central point that emerged from the research is the divergence between real agency (theorized by classical artificial intelligence) and simulated agency (implemented today). The BDI (Belief-Desire-Intention) architecture is based on explicit and persistent mental states: the system believes in a model of the world and desires to achieve priority goals through formal deliberation.

In modern Agentic AI, this process is emulated through Agentic Workflows. In this case, you don't find stable internal mental states, but an iterative loop that turns a user query into a plan, executes it via external tools, and reflects on the results to make any corrections. It is precisely this capacity for reflection (self-reflection) that creates the illusion of proactivity: the system does not want to act, but is programmed to cyclically re-execute the reasoning until the results of the action satisfy the criteria set in the prompt.

Feature	BDI Agents (Symbolic AI)	Agentic AI (LLM-based)
Paradigm	Symbolic, Logical and Deterministic.	Connectionist, Probabilistic and Stochastic.
Mental States	**Persistent**: Beliefs, Desires and Explicit Intentions.	**Transients**: Simulated within the Prompt Context.
Reasoning Loop	BDI Loop (Update, Generate, Filter, Deliberate, Execute).	Agentic Workflow (Planning, Tool Use, Reflection).
Persistence	Exists continuously between reasoning sessions.	Vanishes entirely when the session ends.
Nature of Agency	Possesses independent internal mental states.	Simulate agency through iteration and feedback.
Robustness	Structural and mathematically verifiable.	Functional; depends on the capacity for self-reflection.
Flexibility	Low: Limited to default logic.	High: dynamic adaptation to new contexts.

Table 4: Comparison between Classic BDI Agents and LLM-based Agentic AI - Own elaboration

5.3 Operational Limits and Latency

Despite the potential, the results show non-negligible technical barriers. The price to pay for reasoning is latency. While a deterministic system responds instantly, an agentic AI requires a significant amount of time to process and plan language, making it less suited to critical real-time contexts. Furthermore, these systems are not yet robust by default, but require external monitoring and security measures to prevent anomalous behavior.

6 Conclusions

For this project, a systematic review of the literature focused on the evolution of Artificial Intelligence systems was conducted, analyzing the transition from traditional deterministic paradigms to emerging agentic AI architectures. The goal of the review was to systematically map the current state of development and knowledge on the topic, assessing how the scientific literature is currently addressing the transition to systems capable of simulating their own autonomy and proactivity.

To ensure the project's success, ten papers were analyzed, selected through a quality assessment and selection process. The purpose of the selection process was to select studies that represent the most advanced research directions on the topic. The analyses conducted on the selected papers led to three key conclusions:

- Summary of Results
- The Ontological Gap
- Future Work

6.1 Summary of Results: The Trade-off between Flexibility and Robustness

The systematic review confirms that the adoption of agentic workflows introduces unprecedented operational flexibility, allowing the software to adapt to dynamic scenarios not previously codified. However, the comparative analysis highlights an inherent robustness cost: the probabilistic nature of LLMs replaces the reassuring rigidity of "If-Then" rules with branches of uncertainty and the risk of operational hallucinations. The analyzed literature indicates that, while traditional systems guarantee absolute predictability, Agentic AI requires new functional monitoring paradigms to manage its stochastic nature.

6.2 The Ontological Gap: Simulation vs Possession

A central contribution of this SLR is the clarification of the gap between classical agency and that implemented through linguistic models. While the BDI (Belief-Desire-Intention) architecture describes entities with persistent and explicit mental states, modern Agentic AI is configured as an agency simulator.

Through iterative workflows of Planning, Tool Use and Reflection, the system emulates intelligent behavior which, however, remains transitory and strictly linked to the context of the prompt, disappearing completely at the end of the session. The review therefore emphasizes that Agentic AI should not be confused with an autonomous agent in the ontological sense, but considered an orchestration architecture based on probabilistic reflection loops.

6.3 Threats to Validity

In this section, potential limitations that may have influenced the results of this systematic review are discussed, classified according to methodological standards for SLR.

6.3.1 Construct Validity

The main threat in this category concerns the terminological confusion currently present in the field.
The main problem is the current confusion regarding the terminology to use in the context studied. Since the term Agentic AI is an emerging paradigm, some important studies may have used different terminologies to describe similar concepts, which could have resulted in the possible exclusion of some articles during the search phase. To minimize this potential problem, the search query was defined to include synonyms such as Agentic Workflow through the use of Boolean operators, thus attempting to cover the widest possible range of related terminology.

6.3.2 Internal Validity

This subsection discusses potential errors introduced during the data selection and evaluation process.
Two potential issues have been identified:

- Subjective quality assessment: Assigning a QA score (1, 0.5, 0) based on qualitative criteria, such as architectural detail or experimental validation, can lead to a certain degree of subjective interpretation.
- Manual selection: Although the PRISMA process is very rigorous, the screening phase, based on title and abstract, resulted in the exclusion of many papers, which increases the risk of selection bias.

6.3.3 External Validity

External validity concerns the generalizability of the results obtained.

- Source Limitation: The paper search was performed exclusively on two databases (ACM Digital Library and IEEE Xplore). Although these two are primary sources for computer science, excluding other archives or databases significantly limits the coverage of the search.
- Open Access Filter: For simplicity and greater freedom, we chose to exclude articles not freely accessible. This move may have removed studies published behind paywalls, limiting the sample analyzed to only open access articles.

6.3.4 Conclusion Validity

The limited number of papers considered and analyzed in depth for the project's scope may not fully represent the full complexity of the global agent AI technology landscape. However, using the PRISMA protocol and a minimum quality threshold ensures that the conclusions drawn are based on the most robust scientific evidence currently available in the 2024-2026 period.

6.4 Future Work

From the literature synthesis it clearly emerges that future research directions will have to focus on the development of hybrid systems. These will have to be able to combine the generative power and adaptability of LLMs with the security and verifiability of formal systems, aiming to reduce the latency and risks of behavioral drift that still slow down the adoption of Agentic AI in mission-critical contexts.

References

[1] Aminiranjbar, Z., Tang, J., Wang, Q., Pant, S., Viswanathan, M.: Dawn: Designing distributed agents in a worldwide network. IEEE Access **13**, 138795–138812 (2025). https://doi.org/10.1109/ACCESS.2025.3588425

[2] CTOI: The belief-desire-intention model (2025), `https://ctoi.substack.com/p/the-belief-desire-intention-model`

[3] Haddaway, N.R., Page, M.J., Pritchard, C.C., McGuinness, L.A.: Prisma2020: An r package and shiny app for producing prisma 2020-compliant flow diagrams, with interactivity for optimised digital transparency and open synthesis. Campbell Systematic Reviews **18**(2), cl2.1230 (2022). https://doi.org/10.1002/cl2.1230, `https://journals.sagepub.com/doi/abs/10.1002/cl2.1230`

[4] Jennings, N., Sycara, K., Wooldridge, M.: A roadmap of agent research and development. Autonomous Agents and Multi-Agent Systems **1**, 7–38 (03 1998). https://doi.org/10.1023/A:1010090405266

[5] K A, A., J, D.U., Subramaniam, U.: A systematic literature review on multi-robot task allocation. ACM Comput. Surv. **57**(3) (Nov 2024), `https://doi.org/10.1145/3700591`

[6] Keele, S., et al.: Guidelines for performing systematic literature reviews in software engineering. Tech. rep., Technical report, ver. 2.3 ebse technical report. ebse (2007)

[7] Khamis, A.: Agentic ai systems: Architecture and evaluation using a frictionless parking scenario. IEEE Access **13**, 126052–126069 (2025). https://doi.org/10.1109/ACCESS.2025.3590264

[8] Lollypop Design Studio: Agentic workflows (2025), `https://lollypop.design/blog/2025/august/agentic-workflows/`

[9] Masters, C., Vellanki, A., Shangguan, J., Kultys, B., Gilmore, J., Moore, A., Albrecht, S.: Orchestrating human-ai teams: The manager agent as aunifying research challenge. In: Proceedings of the 2025 7th International Conference on Distributed Artificial Intelligence. p. 91–107. DAI '25, Association for Computing Machinery, New York, NY, USA (2025), `https://doi.org/10.1145/3772429.3772439`

[10] Narasimhan Sampath, A., Thakur, A., Krishnan, S.: Agentic reasoning for social event extrapolation: Integrating knowledge graphs and language models. IEEE Access **13**, 172096–172110 (2025). https://doi.org/10.1109/ACCESS.2025.3612015

[11] Page, M.J., Moher, D., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Stewart, L.A., Thomas, J., Tricco, A.C., Welch, V.A., Whiting, P., McKenzie, J.E.: Prisma 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. BMJ **372** (2021). https://doi.org/10.1136/bmj.n160

[12] Porter, Z., Calinescu, R., Lim, E., Hodge, V., Ryan, P., Burton, S., Habli, I., Lawton, T., McDermid, J., Molloy, J., Monkhouse, H., Morgan, P., Noordhof, P., Paterson, C., Standen, I., Zou, J.: Insyte: A classification framework for traditional to agentic ai systems. ACM Trans. Auton. Adapt. Syst. **20**(3) (Sep 2025), `https://doi.org/10.1145/3760424`

[13] Rao, K., Coviello, G., Mellone, G., De Vita, C.G., Chakradhar, S.: Xpf: Agentic ai system for business workflow automation. In: Proceedings of the 34th International Symposium on High-Performance Parallel and Distributed Computing. HPDC '25, Association for Computing Machinery, New York, NY, USA (2025), `https://doi.org/10.1145/3731545.3743644`

[14] Sucal, V., Jullien, M., Njifenjou, A., Lefèvre, F.: Llm-based agentic workflow on verbal and non-verbal audiovisual perceptions and actions for proactive situated human-robot interactions. In: Proceedings of the 13th International Conference on Human-Agent Interaction. p. 177–187. HAI '25, Association for Computing Machinery, New York, NY, USA (2026), https://doi.org/10.1145/3765766.3765797

[15] Thanh, C.N., Van, S.N., Ngoc, H.N., Duc, C.V., Minh, H.N., Tien, D.N., Thi, H.N.: Automind: Automated insight discovery via multi-agent navigation. IEEE Access **14**, 23936–23955 (2026). https://doi.org/10.1109/ACCESS.2026.3661202

[16] Wang, Z., Li, R., Li, S., Xiang, Y., Wang, H., Zhao, Z., Zhang, H.: Rally: Role-adaptive llm-driven yoked navigation for agentic uav swarms. IEEE Open Journal of Vehicular Technology **6**, 2693–2708 (2025). https://doi.org/10.1109/OJVT.2025.3610852

[17] Williams, D., Craun, M., Le, M.V., Stephen, J., Ahmed, S., Jamjoom, H.: Towards safe agentic ai performance engineering. In: Proceedings of the 4th Workshop on Practical Adoption Challenges of ML for Systems. p. 63–68. PACMI '25, Association for Computing Machinery, New York, NY, USA (2025), https://doi.org/10.1145/3766882.3767179